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The effect of taxonomic, host-dependent features and sample bias on virus host prediction using machine learning and short sequence *k*-mers

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Metaviromic studies of potential emerging infection reservoirs led to discovery of many novel viruses. Since metaviromes contain viruses from target host, its food or other sources, fast and robust approaches are needed to predict hosts of unknown viruses based on their genome data. Four machine learning algorithms (random forest, two gradient boosting machines, support vector machine) were used here to predict the hosts of RNA viruses that infect mammals, insects and plants. The prediction efficiency was largely dependent on the dataset composition. In the more challenging task of predicting hosts of unknown virus genera, median weighted F1-score of 0.79 was achieved using support vector machine and 4-mer frequencies, a notable improvement over baseline methods (median weighted F1-scores 0.68 for the homology-based tBLASTx and 0.72 for ML trained on mono-, di- and trinucleotide frequencies). More complicated features and feature combinations provided worse results. When predicting hosts of short virus sequence fragments quality decreased but using same-length fragments instead of full genomes for training consistently produced an improvement of prediction quality. Therefore, short *k*-mers carry sufficient information to predict hosts of novel RNA virus genera. This algorithm can be useful in rapid analysis of metaviromic data to highlight potential biological threats.

Keywords Virus host prediction, Machine learning, RNA viruses, Nucleotide *k*-mers, Taxonomic bias

Emerging infections are among the key safety concerns for humanity. A majority of novel virus diseases in the last century were a result of zoonotic transfers (single host switch from an animal reservoir to a human or a domestic animal). The urge to understand these events led to a burst in metaviromics studies in possible source hosts, when a total nucleic acid from a sample is sequenced and analyzed for RNA viruses that could potentially switch to humans[1–8]. Such studies yielded a swarm of novel viruses that are in a clear relation to those that infect humans and domestic animals. However, most of metavirome studies were done on non-sterile samples (usually feces) that contain contaminating nucleic acid from food and other sources, and a large number of sequencing reads could not be identified. Some of these sequences had too little homology with known viruses to infer their host. Therefore, it is desirable to be able to predict host class of entirely novel viruses.

It is well known that genomic sequences (both host and viral) are under various constraints at the nucleic acid level. Some of them are well described, most prominently, the dinucleotide bias[9–15], while more may be expected to remain unknown. The mechanistic explanation of genome composition irregularities often lagged the observation. Codon bias was probably the first to be explored, in analogy to that in the cellular RNAs. However, already an early study attributed most of the codon usage bias in viruses to mono- and dinucleotide frequencies[16]. Later, it was shown that dinucleotide bias acts independently of the codon usage bias, and may even affect choice between compatible amino acids, such as leucine and isoleucine, based on the dinucleotide content in their codons[17]. The key dinucleotide known to be avoided in mammalian (and in many vertebrate) virus genomes is CpG, which was recently found to be a target of the Zinc-finger antiviral protein[18–20]. Another

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underrepresented dinucleotide is UpA, a target of RNAse L[11,21–24]. Other dinucleotides may be significantly under- or overrepresented in certain virus genomes; however, the patterns are not consistent, and the mechanistic explanation is lacking. It was reasonable to use such ubiquitous genome features to infer virus hosts.

The possibility of such task was suggested back in 2010[25]; however, using descriptive statistics was not sufficiently robust for this task. Later studies showed that both taxonomic position[26] and *k*-mer content[26–29] can be used to infer virus hosts using machine learning (ML), as reviewed in[30]. One of the pioneering papers showed virus taxonomy signal impact on host prediction at the family level[31]. Linear discriminant analysis was trained on odds ratios of dinucleotides to predict virus hosts and families of animal RNA viruses and demonstrated that dinucleotide composition of virus genomes carries host and virus taxonomy (family level) signals with the second being predominant. Nevertheless, presence of virus host signal in dinucleotide frequencies provided a sufficient background for ML usage. Further development of virus host prediction included expansion of feature sets (nucleotide and amino acid *k*-mer frequencies with different *k*, functional amino acid features) and use of various ML algorithms. In 2018 gradient boosting machines were shown to outperform logistic regression, naive Bayes, *k*-nearest neighbors, support vector machine and tree ensemble methods in the multiclassification task of prediction of 11 host groups[26]. Feature sets included mononucleotide, codon, amino acid frequencies, relative frequencies of dinucleotides and codon pairs, and so-called phylogenetic neighborhood, which reflects whether the virus has close relatives that infect a particular host species. In a recent study[32] this approach was further optimized to predict hosts of RNA viruses infecting vertebrates, invertebrates, plants, fungi, and bacteria. In one of the central studies[28], the support vector machine classifiers with linear kernels were trained on different representations of virus genomes (nucleotide, amino acid, amino acid properties, and protein domains) to solve the task of binary classification of hosts at different taxonomic ranks for viruses from all Baltimore groups. The increase in nucleotide *k*-mer length from 1 to 9 and amino acid *k*-mer length from 1 to 4 led to predictive power growth in multiple binary classification tasks. Besides a general task of large host taxon prediction, classifiers were aimed for prediction of virus-host interactions for specific groups of viruses[33,34]. Noteworthy, deep learning models were also tried for the task of virus host prediction, but mainly for narrow taxa. The results were generally on par with those of ML; however, no universal solution for animal virus host prediction using deep learning was suggested, and it is difficult to compare results because datasets and tasks were very different[35–38].

While the possibility of virus host prediction using short *k*-mers is generally established, the choice of methods and datasets may have a significant effect on the results. Moreover, many studies used simplified datasets or very focused tasks to prove the concept. It is therefore likely that dataset composition and nuances of methodology affected the results. Here, we explore the effect of dataset composition and variations in methodology on virus host prediction in the tasks that are most relevant in terms of fecal virome studies. Machine learning models were compared to homology-based methods (tBLASTx) and Host Taxon Predictor, a ML framework trained on complete virus genomes using a combination of short *k*-mers[27], which includes tasks and tools that were most similar to our approach.

Materials & methods

Data collection

Complete genome sequences ($N = 17482$) of RNA viruses infecting mammals (class Mammalia), insects (class Insecta), and plants (clade *Viridiplantae*) were downloaded from Virus-Host Database (Virus-Host DB) in April, 2023. Virus-Host DB is a curated database of taxonomic links between viruses and their hosts, that covers viruses with complete genomes stored in NCBI/RefSeq and listed in EBI Genomes. The database includes virus lineage and host lineage annotation. According to each genome accession number, amino acid sequences ($N = 694002$) were retrieved.

Sequences of arboviruses (arthropod-borne viruses that alternate vertebrate and arthropod hosts) were excluded, and segmented genomes were concatenated. As families *Picornaviridae*, *Coronaviridae*, and *Caliciviridae* were overrepresented, closely related genomes with identity greater than 92% were excluded. The final dataset consisted of 1363 virus genomes and the corresponding amino acid sequences of coding regions. The final dataset included viruses from 42 families, with 26 families of ssRNA(+) viruses, 10 families of ssRNA(-) viruses, and 6 families of dsRNA viruses (Supplementary Table S1).

Sequence features

To convert each nucleotide or amino acid sequence into a numeric vector, *k*-mer frequencies with sequence length normalization were used. Nucleotide (NT) *k*-mers frequencies were computed with *k* in range from 1 to 7; amino acid (AA) *k*-mer frequencies were computed with *k* in range from 1 to 3 and averaged over all proteins (translated coding sequences) available for the corresponding genome. One of the goals of this study was to investigate the predictive power of different feature combinations. For this purpose, experiments with different combinations of NT and AA *k*-mer frequencies used for training were conducted.

Approaches for train-test split

The dataset was divided into train (67–78% within replicates) and test (22–33%) with the same ratios of host classes (Fig. 1). Several partitions were created to evaluate predictive power of models on previously unknown viruses. Genome sequences were grouped by taxa level (Families, Genera), and the following approaches for train-test split were performed:

1) All families were equally presented in both training and test datasets (hereinafter referred to as "Closely related"; approximate ratios of insect, mammalian and plant hosts were 0.25 : 0.49 : 0.26).
2) Families in the test dataset were not used during training (hereinafter referred to as "Non-overlapping families"; insect, mammalian and plant host approximate ratios were 0.26 : 0.50 : 0.24).

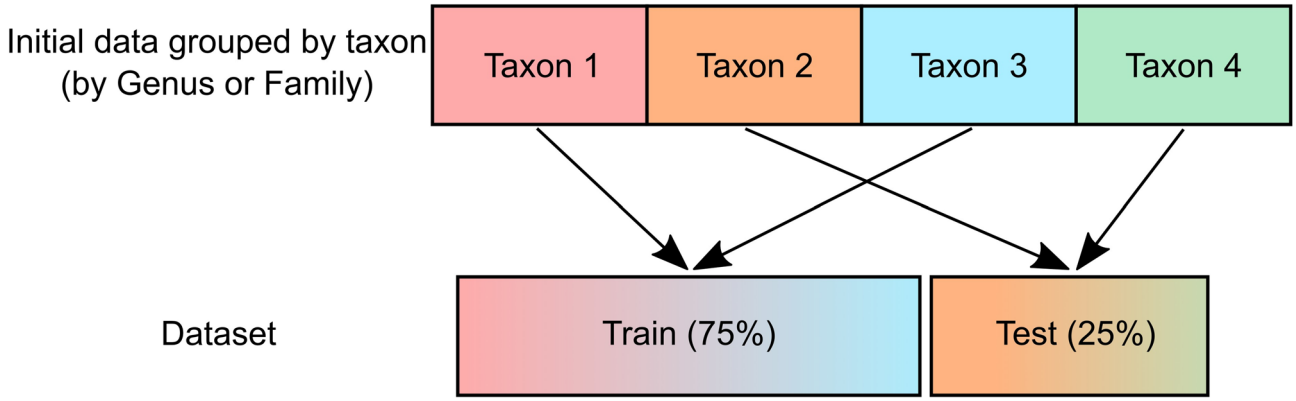

Fig. 1. The scheme of data partitioning into training and test sets based on virus taxa ("Non-overlapping families" and "Non-overlapping genera" approaches). Virus genomes ($N = 1363$) were grouped by taxon (family or genus). Then the sequences were split into training and test datasets maintaining host class ratios so that the taxa of viruses in training and test datasets did not overlap.

3) Genera in the test dataset were absent in the training dataset (hereinafter referred to as "Non-overlapping genera"; insect, mammalian and plant host approximate ratios were 0.25 : 0.49 : 0.26).

Ten randomized iterations were performed for each type of dataset split.

To test the capacity of ML to identify hosts by sequences of single reads or small contigs produced by next generation sequencing, additional datasets of shorter genome fragments were created. In the train-test splits of the complete genome datasets, each genome was divided into non-overlapping subsequences of 400 or 800 nucleotides, then only two fragments per genome were randomly selected.

Machine learning

Four machine learning algorithms were trained: random forest (RF; scikit-learn, Python), gradient boosting machines (LightGBM (LGBM) and XGBoost (XGB); lightgbm and xgboost libraries, Python), and support vector classifier (SVC; scikit-learn, Python). The models' hyperparameters were optimized using cross-validated grid-search over a parameter grid (scikit-learn, Python). To reduce calculation time, ML models were tuned using only for 1 out of 10 partition iterations, for other 9 the same hyperparameters were used. Multiclass classification results for SVC were obtained based on the highest value out of three binary host classification probabilities: insects vs. others, mammals vs. others, plants vs. others. For model evaluation binary and weighted F1-scores were computed (scikit-learn, Python).

Baseline methods

Machine learning models trained in this study were compared with the dummy classifier, homology-based method based on tBLASTx, and a published ML method Host Taxon Predictor[27].

Dummy prediction

The dummy classifier assigns the most frequent host class in the test dataset for all sequences.

Homology-based prediction (tBLASTx)

The BLAST database was created with training sample sequences, and the search of test sequences (queries) against the database was performed. High-scoring Segment Pairs (*max_hsps*) threshold was set to 1. Only five best hits per query (*max_target_seqs*) were selected to reduce the number of tBLASTx results that need to be processed. The length of the seed that initiates an alignment (*word_size*) was set to 4. Hits with alignment length higher than 100 nt were selected. Then for each hit weight was calculated using the following Eq. (1):

$$weight_j = \frac{w_j}{\sum_{i=1}^{k} w_i} \tag{1}$$

where k (total number of hits) $<= 5$, $w_j = pident$, $pident$ is percentage of identical matches.

Hits per query were grouped by host class, and weights were summed for every group. The resulting values were treated as support for each host group. Finally, the host with the highest support was selected as the host for the query. Prediction results were evaluated using weighted F1-score metric.

Host taxon predictor

The second reference benchmark was based on models used in a similar ML-based host prediction framework, the Host Taxon Predictor[27]. Open-source code was used to calculate features according to the original publication. In total, 101 features for every sequence were calculated: nucleic acid type, 4 mononucleotide frequencies, 16 dinucleotide frequencies, 16 relative dinucleotide frequencies, and 64 trinucleotide frequencies. Unfortunately,

models trained to identify plant-, vertebrate-, and arthropod-specific viruses were not published. Therefore, the ML models (logistic regression and SVC) were trained according to the Host Taxon Predictor study, "Materials and Methods" section and open-source code[27,39].

Results

Data collection

Viruses infecting host taxa Mammalia, *Viridiplantae* and Insecta are the most relevant groups in terms of fecal virome studies and include enough virus taxa to explore the capacity of machine learning to predict virus host. Arboviruses (arthropod-borne viruses that infect hosts both mammalian and insect hosts) were excluded from the final dataset as their genomic features are potentially shaped by different hosts. The final data set consisted of 1363 genomes from 42 virus families represented by 160 genera (Supplementary Table S1). Of these viruses, 49.5%, 24.7%, and 25.8% infect mammals, insects, and plants, respectively. Since evolutionary clade *Viridiplantae* is the most variable (includes species that are the most distant from each other), the taxonomy of each host that belong to this clade was investigated. As a result, 351 out of 352 plants represented flowering plants (class *Magnoliopsida)* and one host species belonged to cycads (class *Cycadopsida*, clade *Acrogymnospermae*).

Predicting hosts of known and novel taxa

Metagenomic sequencing can yield genomes of viruses that belong to known taxa, to novel genera and novel families. Identification and prediction of virus hosts in the latter cases can be most challenging. Therefore, three approaches to dividing the dataset into training and test samples were used as detailed above. First, the training and test set were sampled randomly with virus families and hosts equally presented in each set. Random Forest, LightGBM, XGBoost and Support Vector Classifier models were trained to predict virus host based on the frequency of nucleotide *k*-mers in viral genomes (Fig. 2A, Supplementary Fig. S1). Even mononucleotide frequencies were quite good predictors of a virus host (median weighted F1-scores for ten replicates ranged between 0.78 and 0.80 for all four ML models compared to 0.32 for a dummy prediction). Di-, tri- and tetra-nucleotide frequencies were all good predictors of a virus host, with median weighted F1-score values between 0.92 and 0.96. This exceeded the prediction quality of the Host Taxon Predictor used as one reference method (median weighted F1-score equaled 0.89), but was close to the results of tBLASTx (median F1-score equaled 0.93) chosen as another reference. The best median weighted F1-score was achieved by the SVC model using 4-mers. Longer *k*-mers worked well for the SVC model but were inferior for other ML models. Even though not superior over tBLASTx in terms of precision in a simple task of predicting a host of known viruses across huge datasets, pre-trained ML classifiers significantly speed up the calculations: BLAST algorithms perform search through the database for every query sequence to find similar sequences with known hosts, while an ML model, once trained, can predict the host of a new sequence after a simple computation of genome features. In some cases, ML classifiers provided even more accurate prediction compared to the tBLASTx method.

A more realistic and challenging task for host prediction is the case of unknown novel genera and families. When training and test datasets did not include closely related viruses, all methods produced significantly lower prediction quality (Fig. 2B). Mononucleotide frequencies were the least informative for virus host prediction, with median weighted F1-scores between 0.60 and 0.64, which were still higher than the dummy prediction weighted F1-score of 0.29. Longer *k*-mers yielded scores between 0.70 and 0.79 using all four models, with $k=4$ and SVC model providing slightly better results (median weighted F1-score = 0.79). Reference methods, Host Taxon Predictor and tBLASTx, produced median weighted F1-scores of 0.72 and 0.68, respectively. The robustness of prediction (variation between ten training/test sample replicates) fell sharply. Typical difference between the most and the least successful replicates was about 0.20 for non-overlapping genera, compared to <0.05 for closely related virus datasets, which allowed (and, statistically, almost guaranteed) that viruses from a genus are present in both training and test datasets (Fig. 2A). However, even in the worst training/test dataset replicates the host prediction results were significantly better than a dummy prediction. Therefore, ML could predict well a host of novel virus genera using short *k*-mers. Machine learning models (methods reported here and the Host Taxon Predictor) were clearly superior over tBLASTx. Although the latter provided a fair result in half of replicates, the spread of performance scores among dataset replicates was higher (0.39) compared to the other models.

Predicting hosts of novel virus families is yet a more challenging task (Fig. 2C). Mono-nucleotide frequencies were moderately predictive of the virus host, with median weighted F1-score ranging between 0.50 and 0.53 vs. the dummy prediction value of 0.32, and with a very good reproducibility between replicates and ML models. Longer *k*-mers yielded essentially the same median weighted F1-scores, but much higher variation between replicates, from 0.17 (i.e. twice lower than a dummy prediction) to 0.76. Host Taxon Predictor provided essentially the same prediction quality (median weighted F1-score = 0.49) as ML-based approaches suggested here. Therefore, longer *k*-mers can be predictive of a host of unknown families, but this is not robust and is highly dependent on training and test sample composition. tBLASTx was apparently not applicable for identifying hosts of viruses representing unknown families.

The effect of taxonomic richness and training dataset composition on prediction quality

To better understand the reasons for variations in prediction quality across different realizations of the train-test split, we examined the composition of the training dataset for non-overlapping genera strategy (Fig. 2B) in greater detail. Since the primary objective was to maintain the ratio of susceptible hosts while ensuring previously known viruses (virus genera) did not appear in the test sample, the number of taxa (virus families or genera) in the training dataset varied across split realizations. Specifically, the number of unique families and genera in ten train-test split iterations ranged from 17 to 23 and from 32 to 48 respectively. We observed that the F1-score

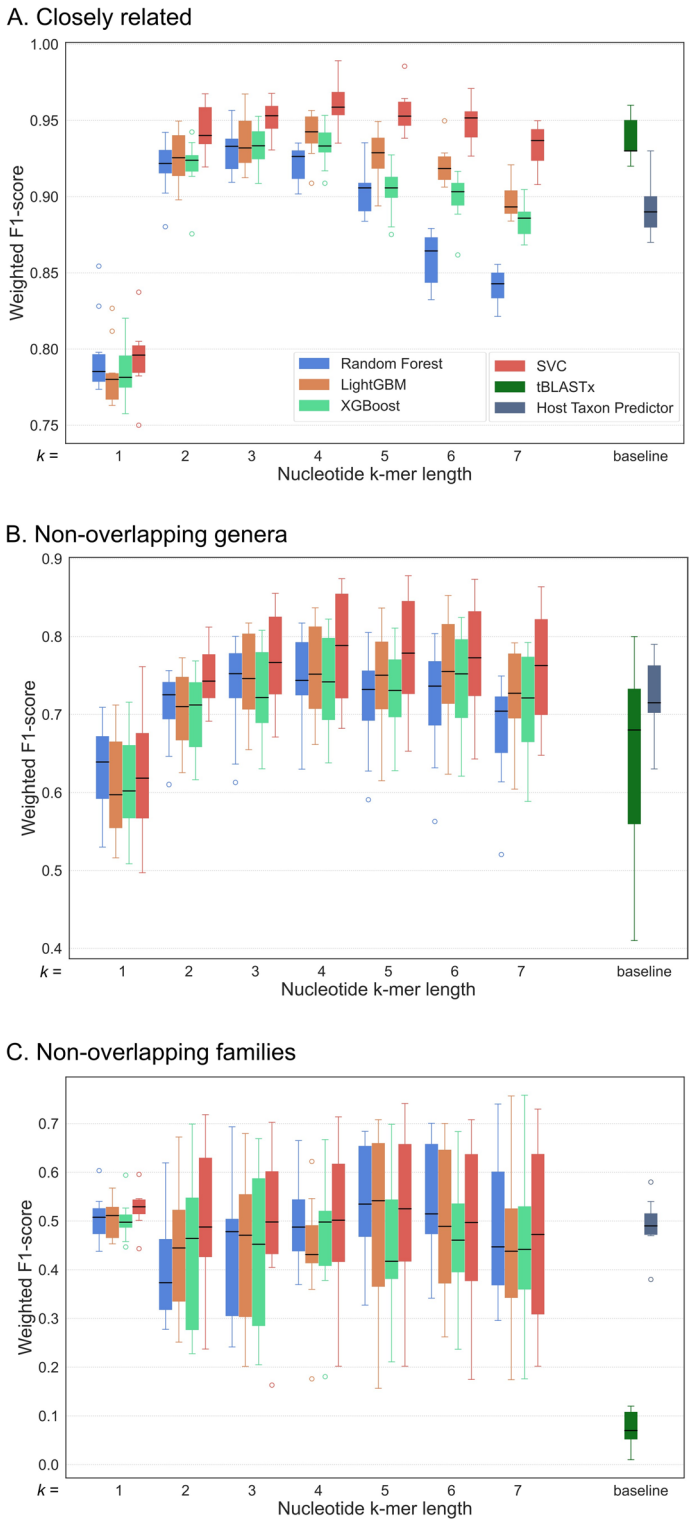

Fig. 2. Weighted F1-score of models trained to predict virus hosts using nucleotide feature sets in various dataset splitting strategies: all families represented in both training and test samples, (closely related) (**A**); genera presented in the test sample were not included in the training dataset (non-overlapping genera) (**B**); families presented in the test sample were not included in the training dataset (non-overlapping families) (**C**). Boxplots indicate distribution of weighted F1-scores for ten train-test split replicates. The models trained in this study (RF - blue, LGBM – orange, XGB – green, and SVC – red) were compared to dummy prediction (black dot line) and two reference methods – homology-based tBLASTx (dark green) and Host Taxon Predictor (dark blue). For Host Taxon Predictor the result of the best model (linear regression or SVC) is shown.

tended to increase with greater taxonomic richness (the number of taxa), though no statistically significant correlation was found between these variables (Supplementary Fig. S2, Pearson's $r = 0.349$, p-value = 0.323).

The observed variation in prediction quality may stem from the classifier's uneven performance across different virus families. Hosts of some families were much more likely to be predicted. The highest prediction probability was observed for *Picornavirales* and other viruses with a ssRNA(+) genome. The top-performing model (SVC trained on nucleotide 4-mers) correctly predicted hosts in fewer than half of the cases for plant viruses of the *Bromoviridae* family and misclassified hosts in nearly half of the cases for viruses of the *Rhabdoviridae* family – a group that includes viruses infecting all three host classes (Fig. 3). It may be tempting to speculate that *Rhabdoviridae*, a unique virus group that infects all three host classes, did not adapt its nucleotide content to any, but the current data do not allow such conclusion. Additionally, lower accuracy was observed for mammalian viruses of the *Paramyxoviridae* family and plant viruses of the *Secoviridae* family.

Nucleotide and amino acid feature sets

Predicting hosts of novel virus genera appeared a feasible task even though the taxonomic signal was largely lacking. Incorporating a range of complementary genome features – such as nucleotide, amino acid (or functional amino acid class) *k*-mer frequencies – has been shown to improve accuracy in virus host prediction tasks, as demonstrated in previous studies[28]. To further improve host prediction quality in this task, nucleotide-based features were compared to amino acid features and their combinations (Fig. 4, Supplementary Fig. S3). Combinations of NT feature sets in the range between 1 and 7 or arbitrary combinations of NT feature sets (data not shown) did not lead to any prediction quality gain. The most informative features varied across classifiers trained on the same feature set but different train-test splits. Despite variability, certain features consistently appeared in the top 10 most important across all models trained on the same feature set (Fig. 5). *K*-mers with a known mechanism of variation among virus taxa (CpG, a target of ZAP, UpA, a target of RNAse L, ApC/ApT, a target and a result of APOBEC editing in SARS-CoV-2) were among the important features, but not on the leading positions. Moreover, dinucleotide features were not found at all when comparing importance of all possible *k*-mers with k between 1 and 7.

Classifier trained using only AA *k*-mers showed poor performance compared to models trained on NT feature sets, although even the least successful replicates were better than the dummy prediction (weighted F1-score = 0.29). Increasing amino acid *k*-mer length reduced the weighted F1-score. A combination of feature sets from different sequence representations (AA and NT) did not differ significantly from the corresponding NT feature sets.

Principal component analysis (PCA) plots revealed no clear separation between viruses infecting different host classes in the space of nucleotide or amino acid *k*-mers frequencies (Supplementary Fig. S4); however, those belonging to families with fewer classification errors (e.g., *Peribunyaviridae*, *Coronaviridae*) exhibited more defined clustering.

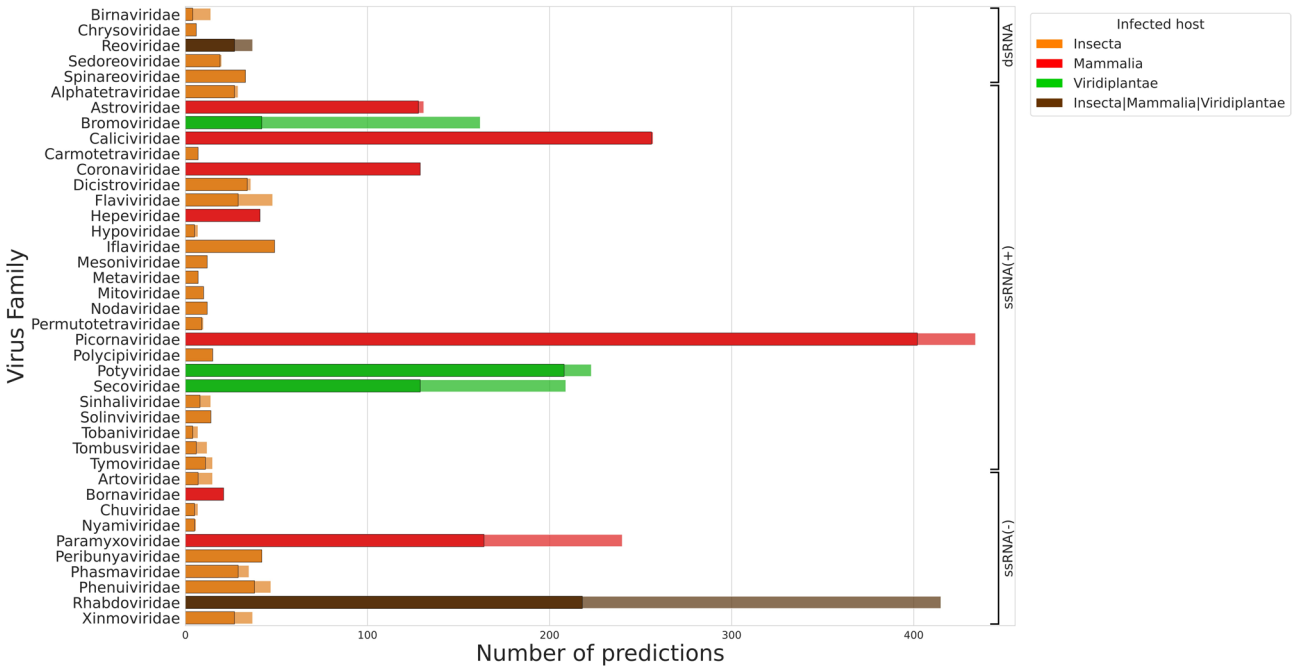

Fig. 3. Accuracy of host predictions across virus families. Results show performance of the SVC classifier trained on nucleotide 4-mer frequencies in non-overlapping genera train-test split scheme, with bars representing: total predictions aggregated across 10 iterations (full height) and correct predictions (darker portion). Virus families are grouped by genome type (ssRNA(+), ssRNA(-), dsRNA) and colored by susceptible host types.

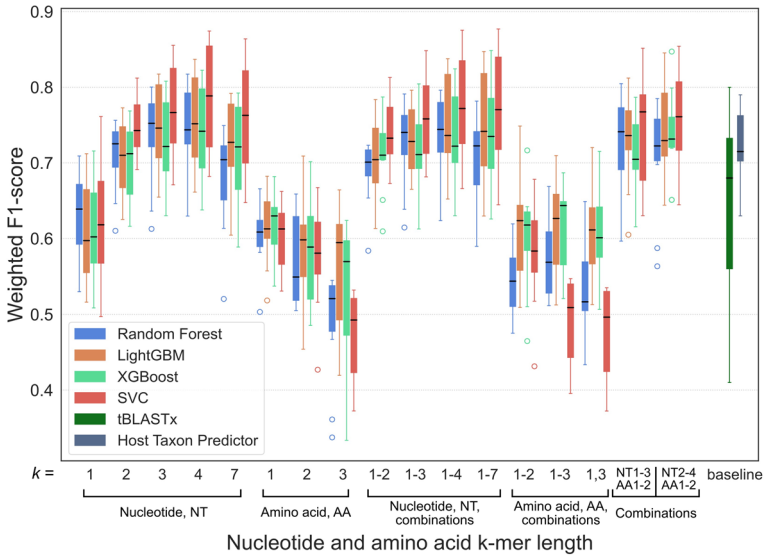

Fig. 4. Weighted F1-score of models trained to predict virus hosts using nucleotide, amino acid feature sets and their combinations. Genera presented in the test dataset were not used during training (non-overlapping genera). Boxplots indicate distribution of weighted F1-scores for ten train-test split replicates. The models trained in this study (RF – blue, LGBM – orange, XGB – green, and SVC – red) were compared to two reference methods – homology-based method (tBLASTx, dark green) and Host Taxon Predictor (dark blue). For Host Taxon Predictor the result of best model (linear regression or SVC) is shown.

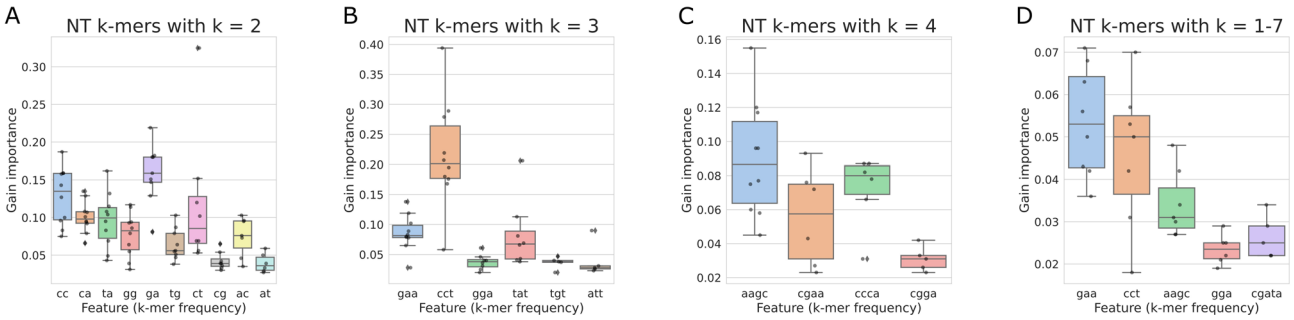

Fig. 5. Feature importance (Gini impurity index) for 10 XGBoost classifiers, each trained on one of the train-test split implementations, using nucleotide frequencies of varying lengths. Panels show results for dinucleotides (**A**), 3-mers (**B**), 4-mers (**C**), and combined *k*-mers ($k = 1–7$) (**D**). Only features ranked among the top ten in more than five models are displayed.

In the amino acid *k*-mer space, data points were generally densely clustered and viruses infecting distinct host classes were indistinguishable, although a few well-defined groups could be observed. These corresponded to plant-infecting viruses of the *Potyviridae* family and mammalian-infecting viruses of the *Picornaviridae* family.

While classifiers trained on amino acid *k*-mers showed overall worse prediction performance, they outperformed nucleotide-based models for 25 out of 178 virus genera (Supplementary Fig. S5). Although training on amino acid features improved prediction for certain Insecta-infecting genera (e.g. *Entomobirnavirus*, *Culicidavirus*, *Gambievirus*, *Jonvirus*, *Mobuvirus*, *Peropuvirus*), the accuracy of predicting viruses infecting insects for the model remained substantially weaker than nucleotide-based predictions. F1-score 95% confidence intervals for Insecta hosts were of [0.651, 0.788] for the best nucleotide model versus [0.521, 0.630] for amino acid models.

Notably, 21 of the 25 genera that benefited from using AA *k*-mers have negative-sense RNA genomes, with 13 concentrated in just two families: *Paramyxoviridae* (6 genera) and *Rhabdoviridae* (7 genera). One could speculate that their genomes are packed with nucleoprotein and this less exposed to cytoplasmic factors affecting sequence content, while they are known to contain several highly conserved amino acid motifs. A modest but consistent improvement in prediction accuracy was observed for plant-infecting positive-sense RNA viruses, particularly in the *Cucumovirus* and *Ilarvirus* genera (family *Bromoviridae*).

Application of host prediction methods to simulated next generation sequencing (NGS) data

The task of host prediction is relevant for the analysis of raw sequencing reads and contigs. Genetic data acquired from metagenomic analysis is represented by reads, which are usually assembled into contigs, and rarely into full genomes. Metavirome reads classification implies predicting hosts not only by full genomes but also by short genome fragments. To solve this task, classifiers trained on full genomes were tested on 400 and 800 nucleotide fragments to infer hosts of viruses that belong to genera absent in the training dataset (Fig. 6A, C, Supplementary Fig. S6, S7). The best results were obtained by using nucleotide di- and tri-mers, although the prediction quality was lower than for full genomes. Median weighted F1-scores were 0.56–0.59 for 400 nt fragments and 0.60–0.63 for 800 nt fragments compared to 0.71–0.75 range for full genomes on the same datasets for RF, LGMB and XGB models. The SVC model showed less reliable and generally worse results in this particular task. Longer *k*-mers were much less informative for classification of short genome fragments, especially for 400 nt, yielding lower weighted F1-score and significant variation between replicates. Median weighted F1-scores obtained here were marginally better than those of the Host Taxon Predictor that uses a very similar approach, while tBLASTx demonstrated very poor host prediction using short genome fragments for viruses of previously unseen genera.

Much better host prediction for short genomic fragments could be obtained when classifiers were trained on fragments of the same length as used for testing (Figs. 6B and D and 7, Supplementary Fig. S6, S7). Dinucleotide frequencies provided a very good result, which was only somewhat improved by using nucleotide triplets. The RF models performed worse on longer *k*-mers in this task, in contrast to classifiers trained on full genomes. Other three models yielded median weighted F1-scores between 0.60 and 0.63 using trimers and 400 nucleotide fragments and 0.62–0.67 using 800-nucleotide fragments. This was 0.04 better than the same models trained

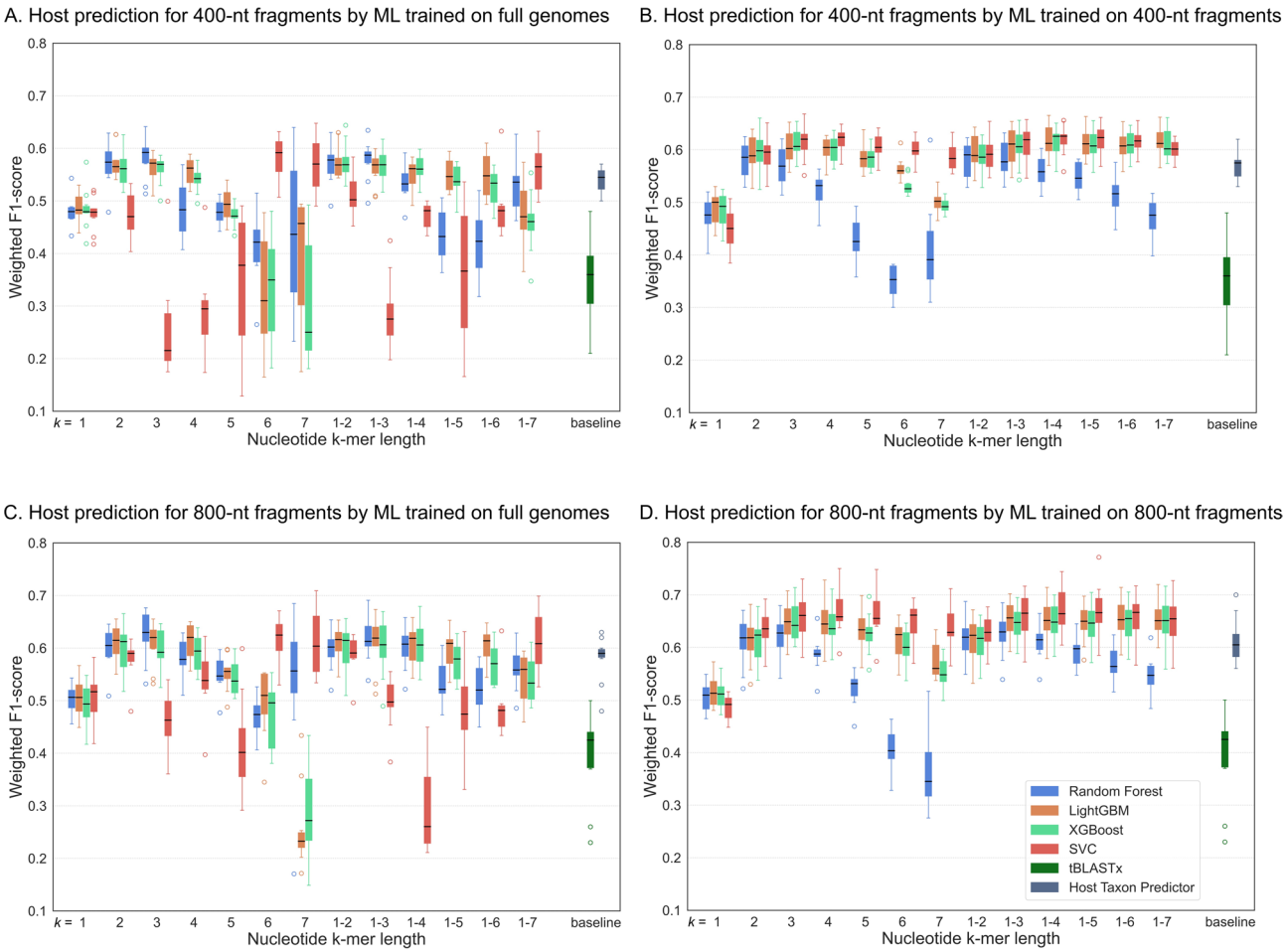

Fig. 6. Weighted F1-score of models tested to predict virus hosts on fragments datasets: classifier trained on full genomes and tested on 400-nt fragments (**A**); classifier trained and tested on the same length 400-nt fragments (**B**); classifier trained on full genomes and tested on 800-nt fragments (**C**); classifier trained and tested on the same length 800-nt fragments (**D**). Boxplots indicate the distribution of weighted F1-scores for ten train-test split replicates. The models trained in this study (RF – blue, LGBM – orange, XGB – green, and SVC – red) were compared to two reference methods – homology-based method (tBLASTx, dark green) and Host Taxon Predictor (dark blue). For Host Taxon Predictor the result of best model (linear regression or SVC) is shown. For tBLASTx a database was created using only the full genome dataset, and a search was performed for the fragments dataset in all cases.

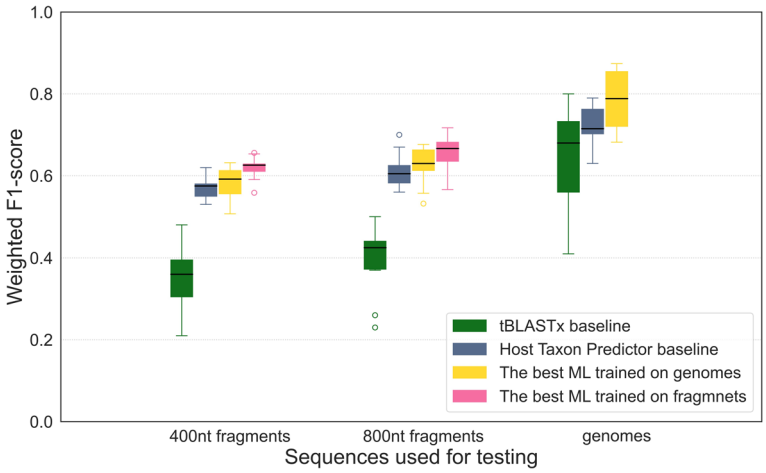

Fig. 7. Effect of training dataset sequence length on host prediction quality. Boxplots indicate distribution of weighted F1-scores for ten train-test split replicates of the ML model that performed best (had the highest median weighted F1-score) for this particular task (Figs. 2B, 4 and 6). The best models trained in this study were compared to two reference methods – homology-based method (tBLASTx, dark green) and Host Taxon Predictor (dark blue). For Host Taxon Predictor the result of the best model (linear regression or SVC) is shown. To obtain tBLASTx-based virus host prediction for short fragments datasets, the database was created using complete genomes from the training dataset and search was performed for sequences from the fragments dataset, similar to approach used in Fig. 6. Host Taxon Predictor was trained on the same kind of data (full genomes or fragments) as tested.

using full genomes (Fig. 7). Longer *k*-mers (5–6 nt in length) were marginally worse than 2–4 nt fragments, but not as appalling as the same 5–6 nt *k*-mers applied for the same task of identifying 400 nt fragments using models trained on full genomes instead (Fig. 6A). Therefore, different *k*-mer length might be preferred for classification of long and short fragments. Most model and feature combinations were better than the HTP, which uses essentially the same features and similar algorithms, and yielded median weighted F1-scores of 0.58 and 0.61 for 400 and 800-nucleotide long fragments, respectively. Using tBLASTx search on a fragments dataset to predict hosts by virus fragments failed completely (data not shown), therefore results of tBLASTx trained on full genomes (same as in Fig. 6A, C) are presented in Fig. 6B, D.

For both fragment sizes datasets LGBM, XGB and SVC models showed very close performance for the most feature sets. RF classifiers, on the contrary, were clearly less capable of predicting virus host using fragments datasets.

With decrease in genome fragments length classifiers drop in median weighted F1-score. Apparently, shorter fragments lose signals inherent in genomes, which could be recognized by classifiers trained on full-length genomes dataset.

Discussion

The task of virus host prediction has two dimensions, analysis algorithms and the underlying phenomena that provide host-specific patterns. *K*-mer content of virus genomes is affected by taxonomic relations (closely related viruses have similar content of short *k*-mers and may share common longer *k*-mers) and convergent evolution (adaptation of the genomic content to more efficient survival and replication in a host). Taxonomic relation of viruses is already a strong predictor of the host[26]. It does not require sophisticated analysis algorithms beyond basic phylogenetic methods, as even tBLASTx yielded fairly good results in this study. However, our results suggest that even in these settings host prediction can be accelerated by using machine learning.

Taxonomic prediction may be helpful in many practical applications, but it is desirable to have capacity to predict hosts of previously unknown virus families or genera. In taxonomic prediction studies, different approaches have been applied to split the dataset into training and test sets. A common way is to divide the dataset by the release date of virus entry in nucleotide databases[40–42], which simulates the case when some virus sequences have not been identified yet. However, such an approach does not guarantee that viruses, even those added to the RefSeq database later, are not closely related to the previously known virus taxa. In[43] datasets with different genetic similarity levels were used for training and testing to show that the suggested model outperforms the existing approaches for novel viruses. Much less attention has been paid to the genetic relatedness of virus sequences used for training and evaluation. *K*-mer feature sets were demonstrated to contain the host signal in holdout experiments, where the phage families were withdrawn from the training dataset and used to test the prediction of host bacteria phylum[28]. However, the effect of taxonomic relation on host prediction has not been studied for RNA viruses infecting eukaryotes. Therefore, the principal approach of training and test sample split used here was chosen to minimize the effect of taxonomic prediction to study host prediction using predominantly host-dependent genomic features. Importantly, using replicates of samples showed that variation from sample bias can exceed the impact of method variation.

Rather unexpectedly, the most important nucleotide content features did not fully correlate with our knowledge of nucleic sequence evolution patterns (Fig. 5). Most of the features important for ML do not have known underlying mechanisms, while motifs that are known to distinguish mammalian viruses, such as CpG depletion, were of moderate importance. Hosts of ssRNA(+) viruses were notably easier to predict. This could be because the genome of ssRNA(-) viruses is better shielded from recognition and modification, or because ssRNA(-) viruses are generally more complex and are hypothetically more likely to have complex mechanisms to block cellular effectors. Host of the plant virus family *Bromoviridae* was consistently difficult to predict with F1-measure below the dummy predictor (Fig. 3). Moreover, when no genus of this family was included into the training sample, the host was almost never predicted (data not shown). This suggests that this family has different (if not opposite) specific composition features compared to other plant viruses.

The key host-dependent (or convergent) feature of virus genomes is the dinucleotide bias. It is conserved within a genus, it may be conserved within a family[31], but can also be highly variable among genera of the same virus family[44,45], which is probably explained by the genus-specific capacity of viruses to counter the corresponding non-self recognition mechanisms of the host cell. Therefore, dinucleotide bias reflects avoidance of host defenses, but also carries taxonomic information, because viruses of the same taxon have similar dinucleotide content[31]. It is likely that a very high quality of host prediction (median weighted F1-score over 0.90) for virus genomes that were randomly distributed into training and test datasets while maintaining the virus family ratio (Fig. 2A, Supplementary Fig. S1) comes largely from a taxonomic prediction, because viruses of the same genus have similar *k*-mer content and infect the same class of hosts. According to theoretical data on the major influence of the dinucleotide content on codon usage[17], using *k*-mers longer than two nucleotides had a limited effect on the prediction quality. When a non-overlapping genera dataset was used for training and testing of ML algorithms (simulating host prediction for novel genera), prediction was good (median weighted F1-score > 0.70), but less robust (as low as 0.59 in some replicates). This result is similar to that observed for bacteriophage host prediction using closely related and non-overlapping taxa[28] and is probably close to the true feature-based host prediction in absence of the taxonomic prediction. *K*-mers longer than two nucleotides provided slightly better results with all ML algorithms, suggesting that there is also an effect of codon usage bias or of certain yet unknown genomic features beyond the dinucleotide content that are host-dependent or conserved at a family level.

Longer *k*-mers (> 4 nt) provided less reliable results. The drop of prediction quality with increasing *k*-mer length is a well-known fundamental limitation of using *k*-mer features. With the increase of length, the probability of observing any specific *k*-mer becomes very small, and most *k*-mers simply do not appear in a training dataset[46]. Thus, rare features involving longer *k*-mers (especially host-specific ones) could not be reliably detected on the background of variation between the sample replicates. The method for more robust estimates of *k*-mer frequencies has been proposed for SVM[46] and was successfully applied for prediction of regulatory sequences[47]. In this study, we attempted to use this approach[48], however, it did not improve the prediction. The *k*-mer length that provides better prediction differs depending on the classification purpose. In[28] and[49], $k >= 6$ was superior to $k = 4$ for host prediction of phages. For binary host classification of eukaryotic viruses, nucleotide *k*-mers longer 4 were superior for RNA viruses from all Baltimore groups[28]. Here, *k*-mers between 2 and 4 were generally good, but notable variation of preferable *k*-mer lengths was observed depending on a task. Therefore, host prediction may benefit from different *k*-mer length depending on biological objects or the number of classes (binary or multiple).

In the case of non-overlapping families, the best prediction was provided by mononucleotide frequencies, while using longer *k*-mers did not consistently improve this result. In cellular genomes, GC content can be affected by various factors[50,51]. Most viruses (in our dataset and in general) have single-stranded genomes. Consequently, G-C (A-T) frequencies are unlinked, and their profile can be more complicated than in cellular genomes. Little is known of factors that affect nucleotide content in virus genomes, but the fact that they provide a notable level of host prediction in any dataset setup of ML algorithms, including non-overlapping virus families, suggests that there are some general patterns of mononucleotide usage.

In addition to nucleotide sequence *k*-mers, informative sequence attributes can include amino acid sequence *k*-mers, *k*-mers of amino acid properties (i.e. simplified amino acid sequences) and protein domains[28]. Protein based features were not generally informative for host prediction of "unseen" viruses simulated as holdout families for phages[28] and here as non-overlapping genera for eukaryote viruses and did not complement the performance of nucleotide-based host prediction (Fig. 4, Supplementary S3). This could be due to the lack of common amino acid-based host-specific signatures, or due to too many features that exceeded the capacity of ML algorithms (for example, for nucleotide and amino acid sequence tri-mers 4^3 << 20^3). It is also noteworthy that combinations of *k*-mers did not produce better prediction scores than isolated features (Fig. 4). Feature combinations have been utilized in the Host Taxon Predictor tool that was used as a reference here. However, single nucleotide sequence feature sets (dimers or trimers or tetramers) performed better than the Host Taxon Predictor trained and tested on the same datasets and about equally well as combinations of these features tried here.

Comparing four ML models in different variations of host prediction shows that they perform comparably well in simple tasks, such as inferring hosts of known viruses. However, in more challenging tasks their performance may be unpredictable. SVC was superior to tree-based ML algorithms in full-genome applications. It provided highly variable and generally worse results when trained on full genomes and used to predict hosts on short fragments (Fig. 6A, C, Supplementary Fig. S6, S7), but fully matched other algorithms when trained in short fragments (Fig. 6B, D, Supplementary Fig. S6, S7). Random forest, on the contrary, performed marginally better than gradient boosting machines (XGB and LGBM) when trained on full genomes and used to predict hosts on short fragments, but failed compared to other methods when trained in short fragments. Thus, it was not possible to suggest a single best ML model to predict virus hosts, and they should be tested thoroughly on samples that are as close to intended tasks as possible. The variation of weighted F1-scores between sample

replicates in this study was relatively small in simple tasks ("closely related", i.e. largely a taxonomic prediction) but increased up to 0.20–0.30 in more challenging datasets. Therefore, estimating statistical significance was not attempted. High variation of prediction quality between the dataset replicates underscores the necessity of exploring the effect of dataset bias on the prediction quality, which in many cases exceeds a minor effect of distinct features or ML models.

Comparing efficiency of ML classifiers suggested here to previous studies was even more challenging. First, the interpretation of the host prediction task and the dataset design varies between the publications. Second, for pre-trained classifiers from other studies it was not possible to rule out data leakage (the training dataset is not available, and same or very similar sequences might be present in the original training sample and in a test sample used when replicating the results). Then, some studies did not present enough dataset replicates; therefore, the robustness of the results was not clear. Furthermore, different studies use various efficiency measures (AUC, accuracy, precision and recall, F1-score).

Even though the SVC algorithm performed very well in most applications, it has its own limitations. When the number of dimensions (number of features) is much greater than the number of objects, the method is likely to provide poor performance. Theoretically, SVC classifiers should fail with $k > 5$ ($N = 1363 << 4^6$) when training on full genomes and with $k > 6$ ($N = 2726 << 4^7$) when training on fragments. In this study, the quality drop was small when using SVC with $k = 7$. One approach to address this problem is using feature selection techniques, which were not applied in this study. Previously, feature selection techniques were shown to have an ambiguous effect on predictions based on genomic data[27], therefore this task goes beyond the scope of this study.

Besides basic implications such as finding host-dependent genomic features in viruses, host prediction has potential usage in biosafety to identify potentially dangerous viruses. ML-based prediction allows rapid analysis of NGS data "on the fly". Our results, as well as previous studies, suggest that ML can provide prediction quality comparable to BLAST on known viruses and much better on novel viruses, while being much faster. However, the prediction quality is known to decrease with sequence length[27]. Indeed, in our sample prediction quality (weighted F1-score) of classifiers trained on full genomes dropped by about 0.15 and became much more variable (Figs. 4 and 6), almost failing in some dataset combinations. An important observation made in this work is that a ML training dataset should be matched to the intended application, which agrees with previously shown results in the task of identifying bacterial viruses from metagenomic data[41]. Machine learning classifiers trained on short sequences were consistently better at predicting hosts of similar length fragments. An average weighted F1-score gain was about 0.04, but more importantly, prediction was much more robust (Figs. 6 and 7). Also, on short fragments prediction quality dropped faster with increasing *k*-mer length. It is noteworthy that in this task (train on short – predict short) SVC provided more robust results with most *k*-mer lengths, while tree-based algorithms were much less reliable on longer *k*-mers ($k > 3$).

A limitation of this study is that classifiers presented here have not been tested on non-virus sequences. The study artificially simplified a real-life task of virus identification by assuming that only virus sequences are tested, which implies an additional preliminary sequence classification. Our aim was to explore the theoretical capacity of the approach, but it might be premature to suggest it for actual preparedness applications, and additional studies are needed to provide consequential or simultaneous segregation of viral and non-viral sequences. Also, it was shown that increasing the number of predicted hosts leads to decreasing prediction quality[35], which might be further aggravated by unbalanced representation of host classes in virus genome databases.

Conclusions

The burst of metaviromic studies of emerging infection reservoirs led to increase of virus genomic data and the discovery of novel viruses. Since metaviromes contain viruses from different sources (target organism host, its food or other sources), new fast and robust approaches are needed to predict hosts of viruses based on their genome data. Machine learning methods have been applied to infer hosts of viruses in specific tasks, and both host signal and phylogenetic relatedness of viruses were shown to contribute to host prediction. Although homology-based host prediction (BLAST search) may be used for known viruses, it fails completely in the case of "unseen" viruses, while ML retains some predictive power even in the most challenging tasks. Our results also show that in most tasks the dataset composition may have greater effect on host prediction efficiency than the choice of algorithms and features. Nevertheless, it is important to train ML machines on the same kind of data (sequence length, taxonomic composition) as would be targeted. Also, in this study robustness of prediction between train-test dataset replicates appeared not less important than the median F1-scores alone.

Data availability

Source code and data are available at https://github.com/WizardFPS/RNA-Virus-Host-Prediction (accessed on 17 June 2025).

Received: 30 June 2025; Accepted: 21 August 2025
Published online: 27 August 2025

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Acknowledgements
This research was funded by Russian Science Foundation, grant number 22-15-00230-П.

Author contributions
Y.A. developed the concept, performed the initial data collection and preliminary analysis of classifiers' performance. F.P. refined the dataset, implemented strategies for train-test split, conducted computational experiments, wrote software and tested its performance. Y.A. reviewed the software. F.P. prepared figures. A.L. supervised work and assisted with result interpretation. All authors drafted and revised the manuscript. All authors accepted the final version.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-17123-w.

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